



Medical Physics Residency

Rotations, Progress and Competency Card

- ☐ Medical Physics Residency Program Graduation Requirements and Status Section
- ☐ **Resident Orientation (2 weeks)**
- ☐ **External Beam Rotation I: Basic Treatment Planning (2 ½ months)**
- ☐ External Beam Rotation II: Advanced Treatment Planning (3 months)
- ☐ External Beam Rotation III: Beam Data, QA&QC, Radiation Safety (3 months)
- ☐ Rotation IV: Imaging (1 month)
- ☐ Rotation V: Brachytherapy and Special Procedures Rotation (3 months)
- ☐ Rotation VI: GammaKnife, Cyberknife, Brachytherapy (PAH) (3 months)
- ☐ Rotation VII: Clinical Elective (PCAM) (3 months)
- ☐ Rotation VIII: Community Practice (LGH) (2 months)
- ☐ Rotation IX: Proton (3 months)



Orientation and External Beam Rotation I: Basic Treatment Planning

Rotation mentors

Knowledge:.....Practicum.....

Rotation Dates:

Reviews:

- Bi-Weekly Progress Review by the Mentor.
- End of rotation review by the Residency Director.

This assignment sheet outlines the requirements for successful completion of the external beam and basic treatment planning rotation. The resident must complete the knowledge and practical factor items listed below. Satisfactory performance is documented by staff signoff with date and passing of the end of rotation oral exam.

Key:

PD = Residency Program Director
APD = Associate Program Director
CR = Chief Resident
EO = Education Officer
RM = Rotation Mentor
SD = Staff Dosimetrist

General Guidelines:

- Orientation (Weeks 1-2)
 - Important ARIA workspace and/or workflows related to clinical treatment planning: Care Path, Contouring, External Beam Planning, Image registration, Documents, Billing, Import/Export
 - Linac and proton treatment console and hand pendant operations
 - Patient simulation (including 4DCT simulation) and treatment set-up
 - Patient-specific quality assurance (PSQA)

- In-vivo dosimetry (IVD) measurements
- Timeline for clinical duty onboarding & credentialing
 - PSQA
 - Weeks 1 – 2: Observation and read procedure
 - Weeks 2 – 3: Practice PSQA (with library case or clinical case under supervision)
 - Week 3 – 4: **Pass Proton/Linac Operation and PSQA Competency Exam**
 - IVD for TBI, TSET, and Cardiac Device protocols
 - Weeks 1 – 3 : Observe and/or perform IVD
 - Week 4: **PGY-1s start undertake IVD assignments**
 - 3D Planning
 - Week 3: Observation
 - Week 4 – End-of-rotation: Practice library plan and undertake clinical planning until end-of-rotation
 - Linac Team Room
 - Week 3 – End-of-rotation:
 - Initial Plan Review
 - Weekly Plan Review
 - SBRT analysis, Special Physics Consults
 - Shadow POW addressing machine issues
 - Share 4DCT responsibility with Planning Support Physicists
 - Linac Monthly QA
 - Month 1 – 2: Observation / Hands-on practice
 - Site Specific Chart Rounds
 - Week 5 – End-of-rotation
- Expectations, Priorities, and Best Practices
 - Journaling
 - Residents will keep a log of all clinical activities throughout residency training
 - Clinical planning
 - Clinical planning takes priority, even on Team days
 - Residents will take up clinical planning only after completing library plans of the same or sufficiently similar site
 - All clinical cases should be reviewed by a dosimetrist prior to physician review
 - Follow Care Path timelines and promptly change Aria task status to “Completed” when task is finished

- Notify lead dosimetrist of any planning delays.
- Communication
 - It is important to maintain timely and efficient communication with the patient care team. Below are a few best practices:
 - Plan-related questions should be addressed to assigned dosimetrist and copy lead dosimetrist and practicum mentor;
 - If you have questions, bundle, escalate and show your homework: Start with searching \\RadOncSharePoint, residency program folder on P:\ drive, residency program channel on MS Teams, and your email. If you cannot find the answer, bundle questions, ask the appropriate person when they are not busy, and show your homework. Bring your questions to a co-resident first, and then escalate if they do not have the answer.
 - Plan delays and deviation from care path should be communicated to lead dosimetrist at PCAM, and to site physicist for network clinics;
 - Resident physicians can be included on key communications for educational purposes;
 - Attending physicians should only be included after checking with assigned dosimetrist, lead dosimetrist/site physicist, or team physicist.

Essential Reading:

- G.C. Bentel, Radiation Therapy Planning, 2nd Edition, McGraw Hill, 1996.
- F.M. Khan and J.P. Gibbons, The Physics of Radiation Therapy, 5th Edition, Lippincott Williams & Wilkins, 2014.
 - Chapters 5, 9, 10, 11, 12, 13, 14, 18, 19
- F.M. Khan, J.P. Gibbons, P.W. Spertudo, Treatment Planning in Radiation Oncology, 4th Edition
 - Chapters 3, 4, 6, 11, 17
- The Modern Technology of Radiation Oncology, Editor J. Van Dyk, Medical Physics Publishing, Copyright 2005.
- AAPM TG 29, “Physical aspects of total and half body photon irradiation”
- AAPM TG 30, “Total skin electron therapy technique and dosimetry”
- AAPM TG 218, “Tolerance limits and methodologies for IMRT measurement based verification QA”
- AAPM TG 40 “Comprehensive QA for radiation Oncology”
- AAPM TG 142 “Quality assurance of medical accelerators”

- AAPM TG 179 “Quality assurance for image-guided radiation therapy utilizing CT-based technologies”

Supplemental Reading:

- Washington and Leaver, Principles and Practice of Radiation Therapy, 4th Edition, Mosby, 2015
- E. D. Halperin, C. A. Perez, L. W. Brady, Principles and Practice of Radiation Oncology, 6th Edition, Lippincott Williams & Wilkins, 2013
- Van de Geijn J, Fraass BA: Net fractional depth dose: A basis for a unified analytical description of FDD, TAR, TMR and TPR. *Med Phys*, 11:784-793, 1984.
- McShan DL, Fraass BA, Lichter AS: Full integration of the Beam's Eye View concept into clinical treatment planning. *Int J Rad Oncol Biol Phys*, 18:1485-1494, 1990.
- BA Fraass, A Eisbruch: Conformal Therapy, Treatment Planning, Treatment Delivery, and Clinical Results. In Textbook, 2nd Edition, Ed. JE Tepper, 2006.
- AAPM TG 63, “Dosimetric considerations for patients with hip prostheses”
- AAPM TG 53, “Quality assurance for clinical radiotherapy treatment planning”, *Med. Phys.* 25(10), 1998
- ICRU Report 50, “*Prescribing, Recording, and Report Photon Beam Therapy.*”
- ICRU Report 62, “*Prescribing, Recording and Reporting Photon Beam Therapy (Supplement to ICRU Report 50)*”.
- M. Van Herk, P. Remeijer, C. Rasch, and J.V. Lebesque, “The probability of correct target dosage. Dose-population histograms for deriving treatment margins in radiotherapy,” *Int. J. Radiat. Oncol., Biol., Phys.* 47, 1121 – 1135 (2000).
- M. Van Herk, P. Remeijer, and J.V. Lebesque, “Inclusion of geometric uncertainties in treatment plan evaluation,” *Int. J. Radiat. Oncol., Biol., Phys.* 52, 1407 – 1422 ,2002.
- Quantec: Quantitative Analyses of Normal Tissue Effects in the Clinic, *IJROBP*, Vol. 76, Issue 3, Supplement, S1-S160, March 2010
- Evaluation of the gamma dose distribution comparison method. *MedPhys* 30(9):2455-64,2003.
- Additional literature and faculty lectures can be found on the Resident shared drive or as directed by the mentor.

Orientation

Complete Orientation training and understand the expectations of Basic Treatment Planning Rotation

(CR,EO) Signature / Date	
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External Beam Knowledge Factors

Discuss with your mentors the following topics on 3D external beam planning, delivery, and verification.

Demonstrate understanding of Megavoltage MU calibration condition and monitor unit calculation for SSD setup.

(RM) Signature / Date	
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Demonstrate understanding of Megavoltage MU calibration condition and monitor unit calculation for SAD setup.

(RM) Signature / Date	
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Demonstrate understanding of photon and electron depth dose curve in homogeneous and inhomogeneous tissue

(RM) Signature / Date	
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Demonstrate understanding of electron ranges (R_p , R_{80} , R_{90} , and d_{max}) for different energies

(RM) Signature / Date	
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Demonstrate understanding of PDD, TAR, TMR, TPR.

(RM) Signature / Date	
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Demonstrate understanding of scatter factors (S_c , S_p , S_{cp}), off-axis factors, inverse square factors, wedge (physical, dynamic) factors, block & compensator factors

(RM) Signature / Date	
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Demonstrate understanding of dose calculation for irregular fields, points outside the field and points under the block.

(RM) Signature / Date	
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Discuss with your mentors the following topics on special procedures in external beam therapy.

Read and discuss Total Body Irradiation (TBI)

<i>(RM) Signature / Date</i>	
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Read and discuss Total Skin Electron Therapy (TSET)

<i>(RM) Signature / Date</i>	
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For each anatomical site, generate a written notes section containing the following information: Disease locations, typical pathology and progression, typical treatment field arrangement, radiosensitive structures, tolerance doses of radiosensitive structures, and prescription doses. The resident shall maintain a notebook with these and other site-specific notes and discuss it with a rotation mentor.

Whole brain

<i>(RM) Signature / Date</i>	
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Head and neck

<i>(RM) Signature / Date</i>	
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Thorax

<i>(RM) Signature / Date</i>	
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Abdomen

<i>(RM) Signature / Date</i>	
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Pelvis

<i>(RM) Signature / Date</i>	
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Extremities

<i>(RM) Signature / Date</i>	
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Site-Specific Treatment Planning Practical Factors

Demonstrate solo operation (under staff supervision but satisfactorily complete without staff intervention) of Eclipse treatment planning systems to generate treatment plans for each of the following sites. The list below include 7 sites requiring clinical planning experience, and 8 sites requiring only library plan practices. Before taking on a clinical plan, the resident is expected to complete a library plan. A total of 22 plans in total, including clinical and library plans, is to be completed in this rotation. If a certain site is not available for clinical planning, it may be substituted with a library plan. Note that multiple iterations on the same CT dataset count as 1 plan only.

At the conclusion of the 3D Rotation, Residents will begin to take emergency on-call treatment planning duties at PCAM and PPMC facilities.

Whole/partial brain

(RM,SD) Signature / Date

Head and Neck (Library)

(RM,SD) Signature / Date

Lung / Chest

(RM,SD) Signature / Date

Lymphoma / Mantle field (Library)

(RM,SD) Signature / Date

Breast tangents with SCLAV (Library)

(RM,SD) Signature / Date

Breast tangents with IMN (Library)

(RM,SD) Signature / Date

Pelvis (Library) (prostate, GYN, bladder, rectum, lymph nodes)

(RM,SD) Signature / Date

Abdomen (Library) (esophagus, pancreas, stomach, spleen)

(RM,SD) Signature / Date

C-Spine

(RM,SD) Signature / Date

T-Spine

(RM,SD) Signature / Date

L-Spine / Sacrum

(RM,SD) Signature / Date

Humerus / Femur

(RM,SD) Signature / Date

Skin electrons

(RM,SD) Signature / Date

TSET (Library)

(RM,SD) Signature / Date

TBI (Library)

(RM,SD) Signature / Date

Note: Library plans or clinical cases will be performed and equally distributed among the sites indicated above.

External Beam Practical Factors

Utilize RadCalc/PennMU software to perform 2nd check of MUs for an external beam treatment plan

(RM) Signature / Date

Utilize RadCalc/PennMU software to perform MU calculation for an electron treatment

(RM) Signature / Date

Perform hand calculation of MUs for one photon 3D plan and one electron plan using either TrueBeam Data Handbook or the commissioning data for RadCalc

(RM,SD) Signature / Date

Perform an in-vivo measurement for a photon radiotherapy treatment field and any necessary calculations to infer the clinical dose delivered. Report your measurements to the Photon Team.

(RM) Signature / Date

Discuss and demonstrate understanding of emergency on-call procedures, including: how to submit a practice emergency plan, attending the emergency on-call meeting, creating Halcyon plans, and when plan QA is required.

(CR,EO,APD,PD)
Signature / Date

Physicist Quality Assurance: Knowledge Factors

Read and discuss the ACR Technical Standard for the Performance of Radiation Oncology Physics for External Beam Therapy

(RM) Signature / Date	
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Discuss AAPM TG-40/TG-142/TG-179, prior to sign-off on performing linac monthly QA.

(RM) Signature / Date	
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Physicist Quality Assurance: Practical Factors

Shadow monthly QA

Machine (Linac/CT): _____

(RM) Signature / Date	
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IMRT QA Knowledge Factors

Explain when and why IMRT QA is required, and what constitutes “IMRT QA” from reimbursement and from technical perspectives. List the tests and tolerance limits for IMRT QA, in our clinical practice **and** as recommended by in AAPM TG 218.

(RM) Signature / Date	
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IMRT QA Practical Factors

Observe and discuss the IMRT QA process (Delivery Quality Assurance or “DQA”) for a Rapid Arc and IMRT plan through completion and all required sign-offs.

(RM) Signature / Date	
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Pass IMRT/RA QA Competency Test

(RM or Senior Resident) Signature / Date	
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Residency Director (PD, APD) Signature / Date	
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